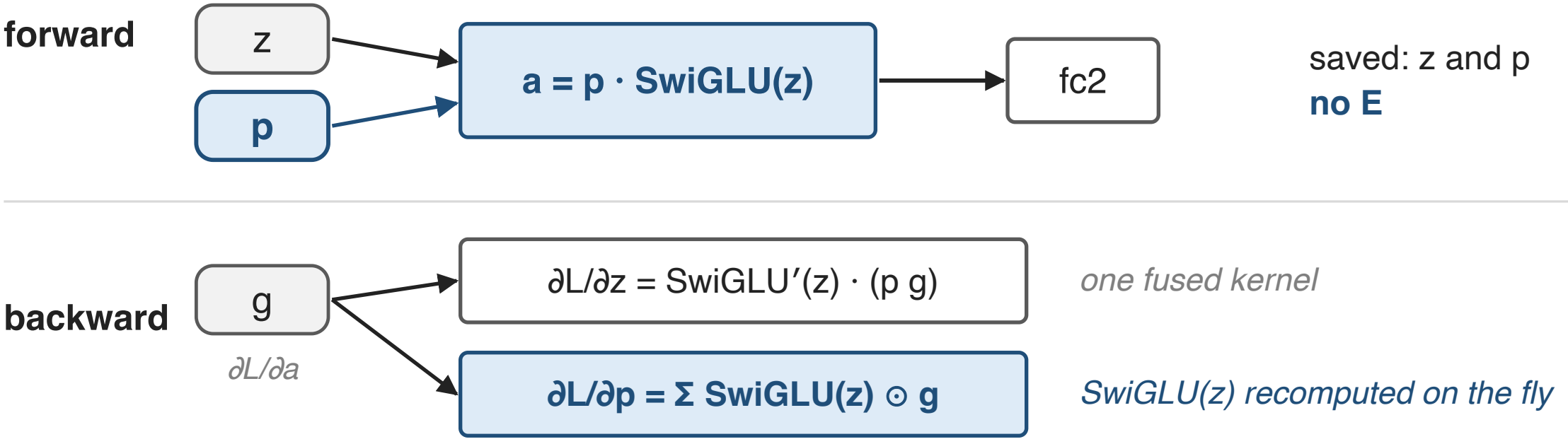

forward
z
p
a = p · SwiGLU(z)
fc2
saved: z and p
no E
backward
g
∂L/∂a
∂L/∂z = SwiGLU′(z) · (p g)
∂L/∂p = Σ SwiGLU(z) ⊙ g
one fused kernel
SwiGLU(z) recomputed on the fly
Σ runs over the row's FFN dimension